



Indian and Korean Foods Image Classification Model using MobileNetV2

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ABSTRACT -- Image classification has become less complicated with deep learning and the availability of larger datasets and computational assets. Food plays a vital role in a human's life as it provides us with different nutrients, and consequently, it is necessary for every individual to keep a watch on their eating habits. Therefore, food classification is important for a healthier lifestyle. Indian cuisine consists of a variety of regional dishes, due to which food image recognition is a necessity. The convolutional neural network is the most popular and extensively used image classification technique in the recent past. In this work, a CNN model and a transfer learning-based MobileNetV2 model are proposed for the purpose of food recognition and classification. To train the classification model, the Korean and Indian food images are collected from the Kaggle dataset. The Korean food dataset consists of 20 food categories with 3467 files. The Indian food dataset consists of 20 food categories with 6269 files. Image classification is performed on diverse food datasets using various transfer learning techniques. Transfer learning is widely used in multi-class image classification, whereby a machine exploits the knowledge gained from a previous task to improve generalisations about another. After experimentation, it was found that the MobileNetV2 model outperformed the CNN model with an accuracy of 70.23% for Korean food images and 69.23% for Indian food images.

Index Terms - Convolution Neural Network, image classification, MobileNetV2, machine learning.

I. INTRODUCTION

Food image classification improves the food experience across the world and also recommends dishes for new eateries and bloggers. Food image classification and recognition help people make the right decisions about their diets. Image classification using a convolutional neural network (CNN) is widely developed for many different purposes. CNN is a neural network that has one or more convolutional layers and is mainly used for image processing, image classification, segmentation, and auto-correcting the data. A convolution is essential for sliding the filter over the inputs. By using CNN, it can reduce the high dimensionality of images without losing the information. CNNs show state-of-the-art performance in fixed-class image recognition. Food image classification is useful in diet management apps for personal health management. Indian foods, in general, have more diverse recipes than Western foods. It is also difficult to define the structure of food because it can be deformed and some of the ingredients can be hidden.

The main objective of this research is to classify the food images using image processing techniques such as (i) digitising the colour image of food images (ii) segmenting those images (iii) extracting features like image size and colour (iv) analysing the results. Classify the type of food and the accuracy percentage. The remaining sections of this paper are described as follows: In Section 2, we represent the literature survey. Section 3 represents the methods and technologies used for identifying the food images. Sections 4, 5, and 6 give a detailed view of the results, discussions, and conclusions, respectively.

II. LITERATURE SURVEY

Chun et al. [1] developed a Korean food image classification model using a public food image dataset and classified images of 20 different types of Korean food using MobileNetV2; they achieved an accuracy of 75.5% on the test set. Arslan et al. [2] focus on fine-grained classification of food items in dietary assessment and monitoring on the UEC food-100 database and report that ResNet50 achieve the highest classification accuracy of 92.69%. Elbassuoni et al. [3] developed a deep learning approach for food recognition based on the NOVA classification system using the DeepNOVA neural network and evaluated their model's

performance against other state-of-the-art deep learning models, showing that DeepNOVA achieved higher accuracy levels compared to other models.

Mandal et al. [4] proposed a deep convolutional generative adversarial network (DCGAN)-based approach for food recognition using partially labelled data and demonstrated the effectiveness of using DCGAN-based models for food recognition tasks and deep learning, highlighting the potential of utilising DCGAN-based models for this task. Hossain et al. [5] proposed a deep learning-based approach to automatically classify fruits for industrial applications. This method has potential applications in the fruit industry, such as fruit sorting and grading, which can increase efficiency and reduce labour costs. Horiguchi et al. [6] proposed a personalized classifier for food image recognition that leverages user-specific dietary information and showed that the personalized classifier significantly outperformed the generic classifier, with an accuracy improvement of up to 10%.

Ciocca et al. [7] proposed pre-trained convolutional neural networks (CNNs) to extract features from food images, apply support vector machines (SVMs) for classification, and evaluate the impact of various factors on the system's performance, such as the number of layers in the CNN, the size of the training set, and the SVM kernel function. Won et al. [8] proposed a novel architecture for fine-grained image recognition that uses a multi-scale convolutional neural network (CNN) with parallel streams of different spatial resolutions to significantly improve the recognition accuracy, especially for fine-grained categories with differences. Xuan et al. [9] proposed a novel approach to fine-grained visual categorization using an evolving convolutional neural network (ECNN), which combines the advantages of both evolutionary algorithms and deep learning techniques to enhance the accuracy of classification for fine-grained images.

III. METHODS

The objective of this study was to develop a classification model for Korean and Indian food images. We collected images from a Korean food and Indian food images dataset to train a model to classify food images. We obtained image data for model training through image pre-processing. The classification model was then trained using the obtained data. Figure 1 shows the training process of the classification model. Each light blue box matches one to one with subsections in the Methods section.

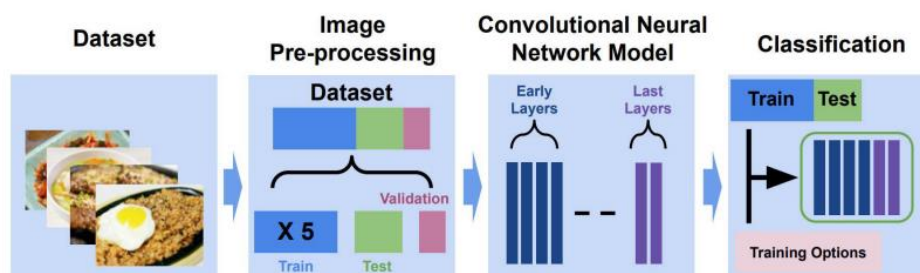


FIGURE 1. Training process of classification models.

The training process of classification models includes (A) Collect Korean food and Indian food images from the Kaggle dataset. (B) Pre-process image to reduce the complexity of the model network, increase the classification accuracy, and prevent overfitting. (C) Load pre-trained feature extraction network and add last classifier layers. (D) Initial training and fine-tuning the trained model.

A. Dataset:

A classification model that automatically classifies images of the visually similar classes of Korean food and Indian food. The performance of a deep learning image classification model that classifies 20 categories of food images using Korean and Indian food images from the Kaggle dataset. The Korean food image dataset consists of 906 images, which are split into 580 images for training, 144 images for testing, and 182 images for validation. The Indian food image dataset consists of 665 images, which are split into 420 images for training, 131 images for testing, and 104 images for validation.

B. Image Pre-Processing

1) Korean Food Dataset

We reduced the complexity of the model network and increased classification accuracy by pre-processing the food images before training the CNNs. We split the image dataset into 580 images for training, 144 images for testing, and 182 images for validation. All the images were resized to 300 pixels by 300 pixels to fit the input of the model.

Bibimbap	Bossam	Bulgogi	Daegae jjim
Dak bokkeum tang	Deonjang chigae	Galbi jjim	Galchijorim
Ganjang gejang	Gimbap	Jjajangmyeon	Jeyuk bokkeum
Kimchi	Nangmyeon	Ojingeo bokkeum	Pajeon
Samgyetang	Samgyupal	Soondae	Soondubu jjigae

TABLE 1. Categories of Korean Food Image Dataset**2) Indian Food Dataset**

We reduced the complexity of the model network and increased classification accuracy by pre-processing the food images before training the CNNs. We split the image dataset into 420 images for training, 131 images for testing, and 104 images for validation. All the images were resized to 300 pixels by 300 pixels to fit the input of the model.

Burger	Chole bhature	Idly	Kulfi
Butter naan	Dal makhani	Jalebi	Masala dosa
Chai	Dhokla	Kaathi rolls	Momos
Chapati	Fried rice	Kadai paneer	Paani puri
Pakode	Pav bhaji	Pizza	Samosa

TABLE 2. Categories of Indian Food Image Dataset**C. Convolutional Neural Network Mode**

We employed a CNN, which is widely used to classify objects in an image, to extract image features. In this study, MobileNetV2 model pre-trained with ImageNet data were used to classify Korean food and Indian food. We connected the pre-trained feature extraction network to the classification model network, which consists of global average pooling layers, fully connected layers, and dropout layers.

D. Classification

Korean food and Indian food image classification modes were trained through two stages: initial training and fine-tuning stages. The aim of the initial training stage was to obtain pre-trained classification models using ImageNet data. During the initial training stage, we trained only the parameters of each classification model. To fix the parameters of the feature extraction network, we set multiple values: the batch size to 128; the training epoch value to 1,000; and the early-stopping value to 50, respectively. We used the Adam optimizer with a 0.001 learning rate and adopted categorical cross-entropy as a loss function.

IV. RESULTS

The different outputs obtained from each stage of the program are displayed in the following figures;

A. Making Predictions on The Test Data

We obtained the performance of the model on a test dataset by displaying a random selection of images along with their predicted and true labels. If the predicted label matches the true label, the title is displayed in green; otherwise, it is displayed in red. It generates a figure consisting of a 3x5 grid of subplots, with each subplot displaying an image from the test dataset along with its true label and predicted label.

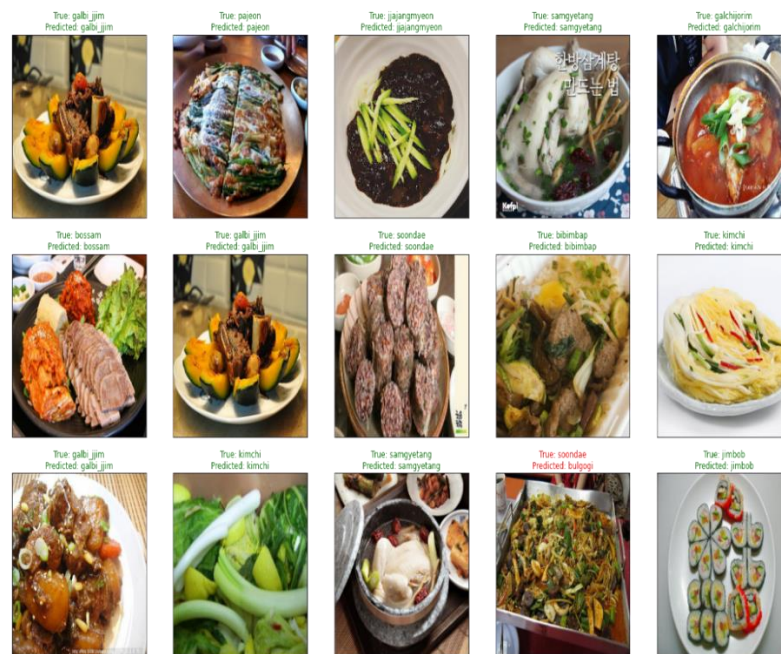


FIGURE 2. Making prediction for Korean Food images

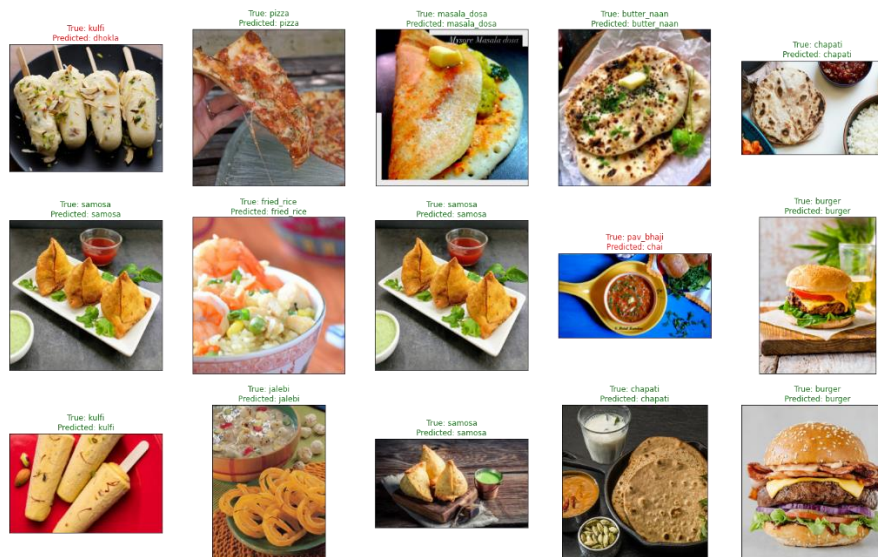


FIGURE 3. Making prediction for Indian Food images

B. Model Evaluation

The graph shows the training accuracy and validation accuracy during the training of the same deep learning model for the Korean food image dataset. The training accuracy measures how well the model predicts the correct output for the training data, while the validation accuracy measures how well the model predicts the correct output for new, unseen data.

The model based on MobileNetV2 outperformed with a test accuracy of 69.23% for Korean food images, a training accuracy of 95.17%, and a validation accuracy of 63.19%. Train accuracy and validation accuracy are also calculated here for a better performance analysis. Their time span from the first to the last epoch is as follows:

Train accuracy: 0.608 to 0.953

Validation accuracy: 0.120 to 0.636

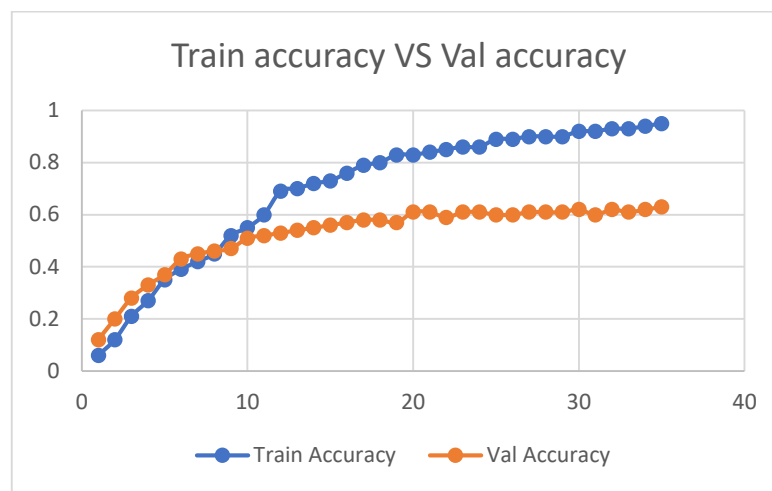


FIGURE 4. Accuracy of Korean Food images

The model based on MobileNetV2 outperformed with a test accuracy of 70.23% for Indian food images, a training accuracy of 97.14%, and a validation accuracy of 69.23%. Train accuracy and validation accuracy are also calculated here for a better performance analysis. Their time span from the first to the last epoch is as follows:

Train accuracy: 0.346 to 0.974

Validation accuracy: 0.157 to 0.696

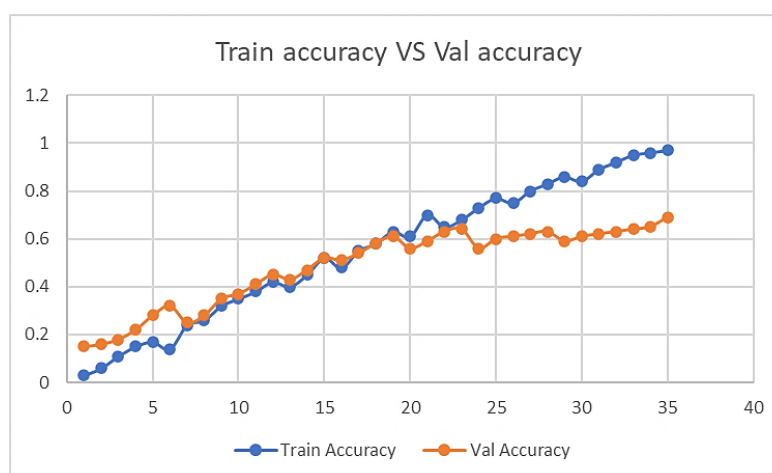


FIGURE 5. Accuracy of Indian Food images

C. Obtained Results

We report the accuracy of each model and the accuracy for each food category. According to the evaluation results, the recall and precision for each food category showed large differences (see Table 3 and Table 4), and the precision and recall of each food type are shown with a representative of each food. The precision, recall, and F1 score are metrics used to evaluate the performance of a classification model. Precision measures how many of the predicted positive examples are actually positive, while recall measures how many of the true positive examples were correctly predicted. The F1 score is the harmonic mean of precision and recall, and it provides a balanced measure of a model's performance. Classification results for the images of Korean foods belonging to bulgogi, soondubu jjigae, and samgyupal had a low recall and precision. Food images in major classes such as daegae jjim, jjajangmyeon, and samgyetang showed the highest recall and precision.

CATEGORY NAME	RECALL	PRECISION	F1-SCORE
Bibimbap	0.57	0.6	0.61
Bossam	0.83	0.62	0.71
Bulgogi	0.428	0.42	0.42
Daegae_jjim	1.0	0.87	0.93
Dak_bokkeum_tang	0.5	0.5	0.5
Deonjang_chigae	0.8	0.61	0.69
Galbi_jjim	0.81	0.69	0.75
Galchijorim	0.6	0.5	0.54
Ganjang_gejang	0.8	0.57	0.66
Gimbap	1.0	0.81	0.89
Jeyuk_bokkeum	0.44	0.66	0.53
Jjajangmyeon	0.85	1.0	0.92
Kimchi	0.66	0.75	0.70
Nangmyeon	0.66	0.71	0.68
Ojingeo_bokkeum	0.83	0.62	0.71
Pajeon	1.0	0.8	0.88
Samgyetang	0.85	0.85	0.85
Samgyupal	0.33	0.5	0.4
Soondae	0.37	0.85	0.52
Soondubu_jjigae	0.36	0.5	0.42

TABLE 3. Classification recall, precision and f1-score of 20 Korean food categories

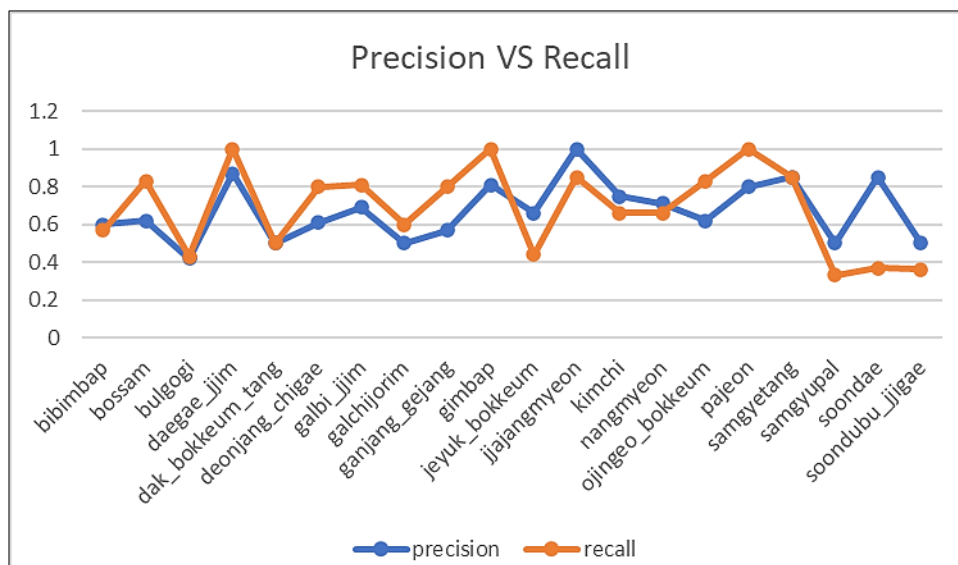


FIGURE 6. Recall and precision of each 20 categories of Korean food data

The classification results for the images of Indian foods belonging to chai, pav bhaji, and kadai paneer had low recall and precision. Food images in major classes such as jelabi, burgers, and pizza showed the highest recall and precision.

CATEGORY NAME	PRECISION	RECALL	F1-SCORE
Burger	1.0	1.0	1.0
Butter_naan	0.75	0.66	0.70
Chai	0.12	1.0	0.22
Chapati	0.75	0.75	0.75
Chole_bhature	0.55	1.0	0.71
Dal_makhani	1.0	0.57	0.72
Dhokla	0.66	0.8	0.72
Fried_rice	0.55	1.0	0.71
Idly	0.75	0.85	0.79
Jalebi	0.85	1.0	0.92
Kaathi_rolls	0.66	0.66	0.66
Kadai_paneer	0.5	0.6	0.54
Kulfi	0.66	0.5	0.57
Masala_dosa	0.75	0.6	0.66
Momos	1.0	0.58	0.73
Paani_puri	0.66	0.5	0.57
Pakode	0.66	0.66	0.66
Pav_bhaji	0.33	0.2	0.25
Pizza	0.83	1.0	0.90
Samosa	0.85	0.66	0.75

TABLE 4. Classification recall, precision and f1-score of 20 Indian food categories

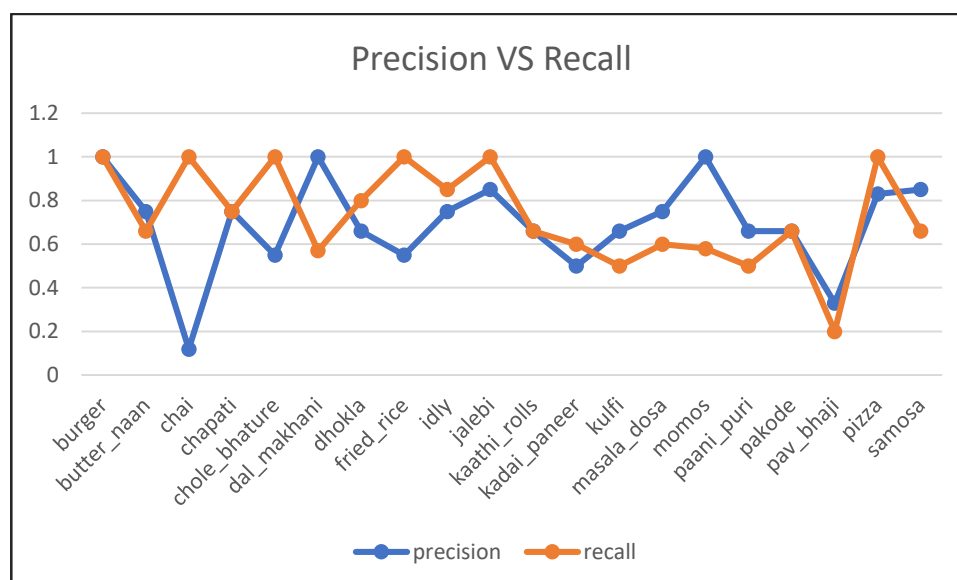


FIGURE 7. Recall and precision of each 20 categories of Indian food data

D. Grad-Cam Visualization

Grad-CAM (gradient-weighted class activation mapping) is a technique that helps to visualise the important regions of an input image used to make predictions. a heatmap that highlights the areas of the image that were most important for the model's decision-making process. The primary purpose of heatmaps is to better visualise the volume of locations and events within a dataset and assist in directing viewers towards areas of the data visualisations that matter most.



FIGURE 8. Grad-Cam visuals Korean food images

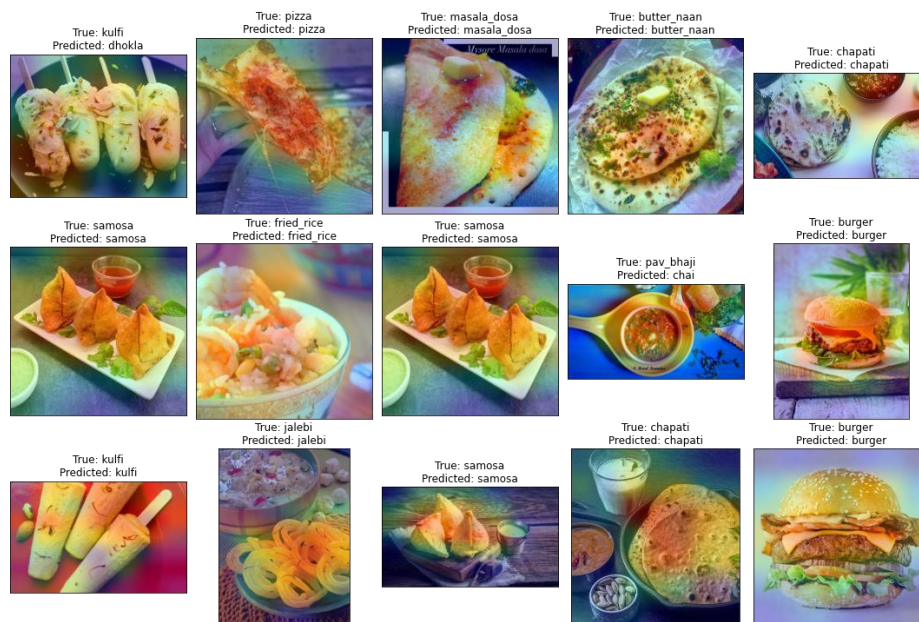


FIGURE 9. Grad-Cam visuals Indian food images

E. Confusion Matrix

The confusion matrix is a matrix used to determine the performance of a classification model for a given set of test data by showing how often each class is predicted correctly or incorrectly. It is a plot comparing predicted labels with true labels. The function first creates a confusion matrix in scikit-learn. If norm is true, it normalises the matrix. It then creates a figure and axes using Matplotlib and plots the matrix with colours representing how 'correct' a class is (darker colour = better).

The test images of pajeon, soondae, bibimbap, and galchijorium were incorrectly predicted as samgyupal, bulgogi, and deonjang chigae, which belonged to a different food category.

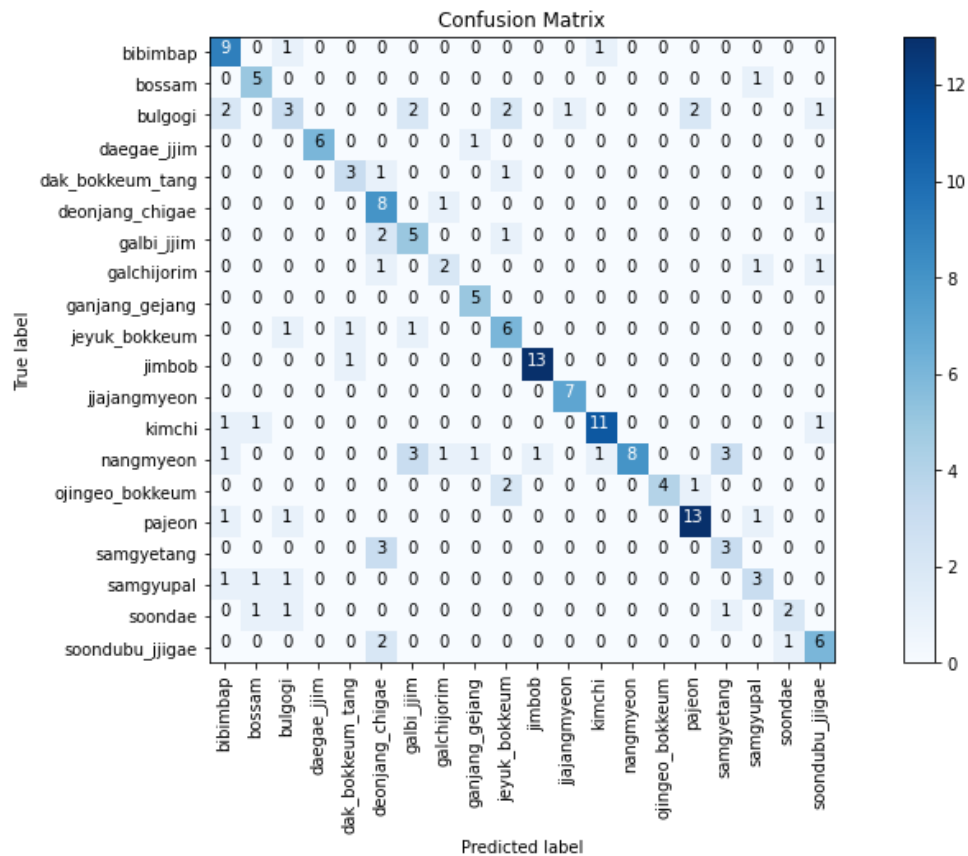


FIGURE 10. Classification result of Korean type food images using model based on MobileNetV2

The test images of kulfi, momos, pav bhaji, and kaathi rolls were incorrectly predicted as chai, kadai paneer, and chapati, which belonged to a different food category.

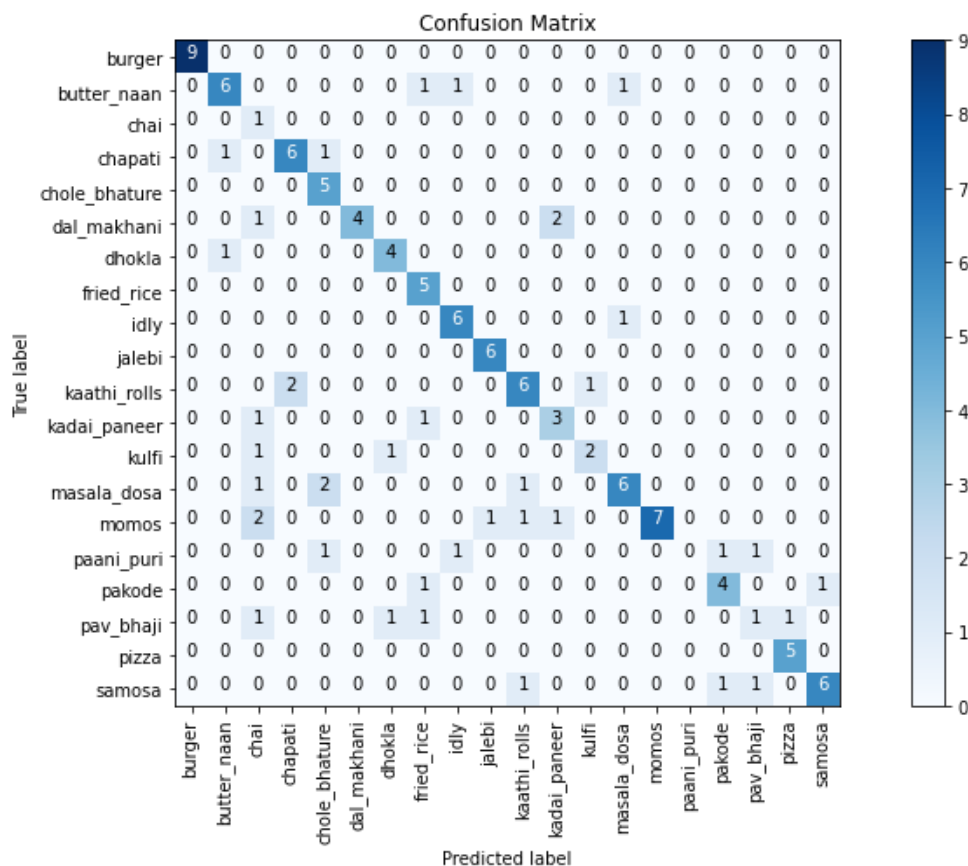


FIGURE 11. Classification result of Indian type food images using model based on MobileNetV2

V. DISCUSSION

We discovered that the classification accuracy of the models was consistent for all the food classes, regardless of the feature extraction networks used. One of the main reasons for the low classification accuracy in certain food image classes was the complex mixing of foods in an image and the high similarities between some image classes. Our findings regarding incorrect predictions within the same category reaffirm the fact that image recognition of Korean food and Indian food is difficult because the images of Korean food and Indian food are complex, and images of foods within the same food class may appear different.

The model based on MobileNetV2 outperformed with a test accuracy of 69.23% for Korean food images, a training accuracy of 95.17%, and a validation accuracy of 63.19%. The model based on MobileNetV2 outperformed with a test accuracy of 70.23% for Indian food images, a training accuracy of 97.14%, and a validation accuracy of 69.23%.

VI. CONCLUSION

We aimed to train a Korean food image classification model. We believe that our model can be used in a system that automatically classifies the type of food that the user has consumed. The existing work was done only with Korean food images, and we have tried the food classification with Indian food images. Users can classify images of similar Korean and Indian foods into 20 categories using the model. The key contribution of this study is that we created a pre-processed dataset for training a Korean and Indian food image classification model. Also, we used two types of datasets to train models that classify Korean food and Indian food images into 20 classes and evaluated the classification accuracy and time required for model training.

REFERENCES

- [1] Chun, Minki, Hyeonhak Jeong, Hyunmin Lee, Taewon Yoo, and Hyunggu Jung. "Development of Korean Food Image Classification Model Using Public Food Image Dataset and Deep Learning Method." *IEEE Access* 10 (2022): 128732-128741.
- [2] Arslan, Berker, Sefer Memiş, Elena Battini Sönmez, and Okan Zafer Batur. "Fine-grained food classification methods on the UEC food-100 database." *IEEE Transactions on Artificial Intelligence* 3, no. 2 (2021): 238-243.
- [3] Elbassuoni, Shady, Hala Ghattas, Jalila El Ati, Zoufekar Shmayssani, Sarah Katerji, Yorgo Zoughbi, Aline Semaan, Christelle Akl, Houda Ben Gharbia, and Sonia Sassi. "DeepNOVA: A Deep Learning NOVA Classifier for Food Images." *IEEE Access* 10 (2022): 128523-128535.
- [4] Mandal, Bappaditya, Niladri B. Puhan, and Avijit Verma. "Deep convolutional generative adversarial network-based food recognition using partially labeled data." *IEEE Sensors Letters* 3, no. 2 (2018): 1-4.
- [5] Hossain, M. Shamim, Muneer Al-Hammadi, and Ghulam Muhammad. "Automatic fruit classification using deep learning for industrial applications." *IEEE transactions on industrial informatics* 15, no. 2 (2018): 1027-1034.
- [6] Horiguchi, Shota, Sosuke Amano, Makoto Ogawa, and Kiyoharu Aizawa. "Personalized classifier for food image recognition." *IEEE Transactions on Multimedia* 20, no. 10 (2018): 2836-2848.
- [7] Ciocca, G., Micali, G. and Napoletano, P., 2020. State recognition of food images using deep features. *IEEE Access*, 8, pp.32003-32017.
- [8] Won, Chee Sun. "Multi-scale CNN for fine-grained image recognition." *IEEE Access* 8 (2020): 116663-116674.
- [9] Xuan, Qi, Haoquan Xiao, Chenbo Fu, and Yi Liu. "Evolving convolutional neural network and its application in fine-grained visual categorization." *IEEE Access* 6 (2018): 31110-31116.
- [10] Tahir, Ghalib Ahmed, and Chu Kiong Loo. "An open-ended continual learning for food recognition using class incremental extreme learning machines." *IEEE Access* 8 (2020): 82328-82346.